

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the	)	Docket No. RM26-4-000
Interstate Transmission System	)	
	)	
	)	

REPLY COMMENTS OF THE LONG ISLAND POWER AUTHORITY

Pursuant to the Federal Energy Regulatory Commission’s (“Commission”) Notice Inviting Comments on the Department of Energy’s proposed advance notice of proposed rulemaking (“Proposed ANOPR”)¹ issued on October 27, 2025² and the Order on Notice on Requests for Extension of Time issued on November 7, 2025,³ in the above captioned proceeding, the Long Island Power Authority (“LIPA”) hereby provides reply comments on the Proposed ANOPR.

I. BACKGROUND

The comments below are technical in nature and reflective of LIPA’s experience operating the Long Island Transmission System. LIPA’s current transmission system is comprised primarily of a 138 kV backbone connecting to 69 kV and lower voltage lines. This system is located in a high population density area and has limited transmission level interconnections to neighboring zones within New York State and external control areas. For this reason, LIPA carefully plans and manages its system to avoid abrupt loss of load scenarios. With potential decommitment and retirement of older, on-island generation plants, LIPA is further emphasizing planning to minimize the risk of system disturbances and adverse interactions between system components, including inverter-based resources and HVDC transmission systems. . Moreover, it is imperative that LIPA,

¹ Ensuring the Timely and Orderly Interconnection of Large Loads, Advance Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 27, 2025).

² Notice Inviting Comments, Docket No. RM26-4-000 (Oct. 27, 2025).

³ Notice Granting Extension of Time, Docket No. RM26-4-000 (Nov. 7, 2025).

as the transmission owner and operator on Long Island, has the flexibility to study technical issues posed by large loads and to set technical interconnection requirements which may not be uniform across the system, or even uniform across the larger New York State bulk transmission system, of which the LIPA system is a part. Simply, there is a critical need for flexibility and adaptation of large load interconnection procedures and standards to ensure the safe and reliable operation of constrained systems, such as the Long Island grid.

II. REPLY COMMENTS

LIPA is hereby replying to matters addressed in initial Proposed ANOPR comments made by stakeholders in this docketed proceeding. LIPA requests that the Commission take into account and incorporate the following recommendations in any future rulemaking related to the treatment of large loads.

A. Large Load Interconnection Studies and Standards Must Take into Account Oscillation Risk.

An area of technical concern that received insufficient attention in the initial comments is the risk posed by the oscillatory behavior of large loads. Certain types of large loads, including data centers, are known to have repeated and large cyclic variations of load demand with periods ranging from a few seconds to a small fraction of a second. For example, artificial intelligence data centers in the training mode of operation are known to exhibit large cyclic load variations. This cyclic stimulus, if it coincides with a “natural frequency” of a grid, can initiate oscillations and cause them to grow to potentially damaging magnitudes.

One consequence of oscillatory behavior of large loads is the torsional oscillations of the shaft systems of large turbine-generators. These oscillations typically range in frequency from 10 – 30 Hz (oscillations per second), which are atypical for a power system. Continued torsional oscillations reduce the mechanical life of the affected generator and, if large enough, can cause

immediate fracture of the shaft, placing critical generation resources out of service for years. Another consequence of oscillatory behavior of large load is generator rotor angle swings across a portion of the bulk transmission system. These oscillations tend to have periods of a few seconds. Continued stimulation of such oscillations, injected at an adverse location, can potentially cause system oscillations to build to the point where the transmission system may separate, potentially leading to wide-scale power blackout.

Proposed Action: We encourage the Commission to incorporate into a proposed rule the following provisions:

- (1) Require the large load interconnection study process to screen for, and assess, the potential for adverse oscillatory behaviors that may require physical (i.e., equipment upgrades) or operational measures to avoid, minimize and mitigate potentially adverse impacts upon the grid; and
- (2) Allow interconnecting transmission owners to adopt general and locationally specific protection measures and equipment standards necessary to limit adverse oscillatory behavior from large loads and co-located loads.

B. Ride Through Requirements Are Critical for Grid Reliability and Are Most Effective Where Transmission Owners and Operators Have the Flexibility to Establish Regionally and Locally Appropriate Standards.

A critical challenge for interconnection of large loads to the bulk power system is how to manage the response of such loads to a system disturbance. Generation-load balances can be disrupted by the tripping of either generation resources or large loads in response to a system disturbance. These events can result in angular instability, over-voltages, and loss of frequency stability. Moreover, such events can snowball, with the load response to a disturbance substantially aggravating the severity of the initial disturbance, particularly if multiple large load facilities trip simultaneously in response to an initiating grid event such as a fault or system separation.

In recent action on North American Electric Reliability Corporation (“NERC”) reliability standards, the Commission recognized the risks associated with system disturbances and changing technologies and has requested further NERC action on frequency and voltage protection standards and ride-through capabilities for inverter based resources.⁴ Recognizing the technological challenges unique to connecting large loads to the transmission system, several initial comments on the Proposed ANOPR included requests for the Commission to adopt threshold applicability requirements based upon specific operating requirements for all loads including, *inter alia*, ride-through capability (i.e., how well the load can remain connected and stable during grid disturbances such as voltage dips, frequency excursions, or short interruptions).⁵

LIPA agrees with the critical need for voltage and frequency protections as well as ride-through standards. However, a one-size-fits-all standard for such measures is short-sighted and increases risk to the grid. Transmission owners and operators must have the flexibility to adapt and adopt prevention and mitigation measures within their interconnection procedures and standards, reflective of their own system’s topology. In order to avoid potential large load tripping behavior on their system, transmission system operators will need to devise comprehensive disturbance ride-through requirements which balance the security needs of the grid with the practical needs of the load facility. Accordingly, frequency and protection measures as well as ride-through standards for large loads cannot be a static, universal standard or study metric. Rather, the transmission owner and operators must have the ability to adapt these standards in a manner that provides an optimal balance of the objectives across all load facilities within its interconnected grid. Requirements

⁴ See, e.g., *Reliability Standards for Frequency and Voltage Protection Settings and Ride-Through for Inverter-Based Resources*, 192 FERC ¶ 61,076 (2025).

⁵ See, e.g., Comments of American Transmission Company at 9, Docket No. RM26-4-000 (Nov. 21, 2025); Comments of Southwest Power Pool Inc. at 18, Docket No. RM26-4-000 (Nov. 21, 2025).

appropriate in one set of circumstances can be too permissive in other circumstances, which compromises grid security. Conversely, requirements that are too restrictive in other cases potentially make large loads economically infeasible. The best balance between these objectives tends to be specific to a particular location in the transmission system based on specific load size and other characteristics. Accordingly, the Commission should ensure that any forthcoming rule provides for regional flexibility on ride through standards in order to meet the needs of a particular system.

Proposed Action: We encourage the Commission to incorporate into a proposed rule the following measures:

- (1) If a large load facility is susceptible to rapid disconnection or load change in response to a system disturbance, then the large load interconnection study process should assess the potential system impact resulting from such abrupt changes; and
- (2) As part of the planning, design and construction of interconnection facilities for large loads, transmission owners should have the right to require load to implement and maintain disturbance ride through capability that is necessary to ensure system stability and security.

C. Increasing the Awareness to Closed Loop Control Interactions

Emerging large loads, such as data centers, are likely to utilize controllable power-electronic devices. The typical control objective, for a receiving load, is to maintain a constant real power load, independent of grid voltage and frequency variations. However, a constant power load inherently provides a negative resistance characteristic. This means that the dampening of oscillations in the bulk system, occurring in the frequency range where the control is responsive, is reduced. If the negative damping characteristic provided by the large load exceeds the natural

positive damping of the grid, oscillations can self-initiate and grow to a disruptive or destructive magnitude. In locations where the transmission system is weak and there are other power-electronic systems such as inverter-based resources (“IBR”), there is potential for the various control systems to interact with each other in an adverse manner across a range of oscillatory frequencies. A modern large load facility connected in the vicinity of IBR, HVDC, FACTS, and other fast-controlled power electronic systems adds further complications and opportunities for adverse interactions.

While not fully researched and documented, there is substantial reason to expect that interconnection of large-load facilities dominated by power electronic loads requires specific attention to potential control interactions with other systems on the grid. The relevance of this issue and the necessary interconnection requirements and restrictions are inherently location-specific based on the strength of the grid and the proximity to potentially interacting systems. System interactions with load devices can also occur even for hybrid facilities having no net power exchange with the bulk transmission system. These aspects should be considered as part of any large load impact analysis. Further, regions should be afforded the flexibility necessary to protect their own systems.

Proposed Action: We encourage the Commission to incorporate into a proposed rule the following measures:

- (1) The large load interconnection study process should assess the potential system impacts resulting from closed loop instability in the system and identify avoidance, minimization and mitigation measures that would reduce the risks of significant adverse impacts to the grid.

- (2) Transmission owners should have the flexibility to identify locationally specific load facility oscillatory dampening characteristics that are necessary for system security and included in the development of the large load's interconnection facilities.

III. CONCLUSION

LIPA appreciates the opportunity to provide reply comments on the questions raised in the Proposed ANOPR and urges the Commission to ensure that transmission owners and operators retain the flexibility necessary to safely operate their systems, while also ensuring that the United States is able to meet the increasing demand for electricity growth.

Respectfully submitted,

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